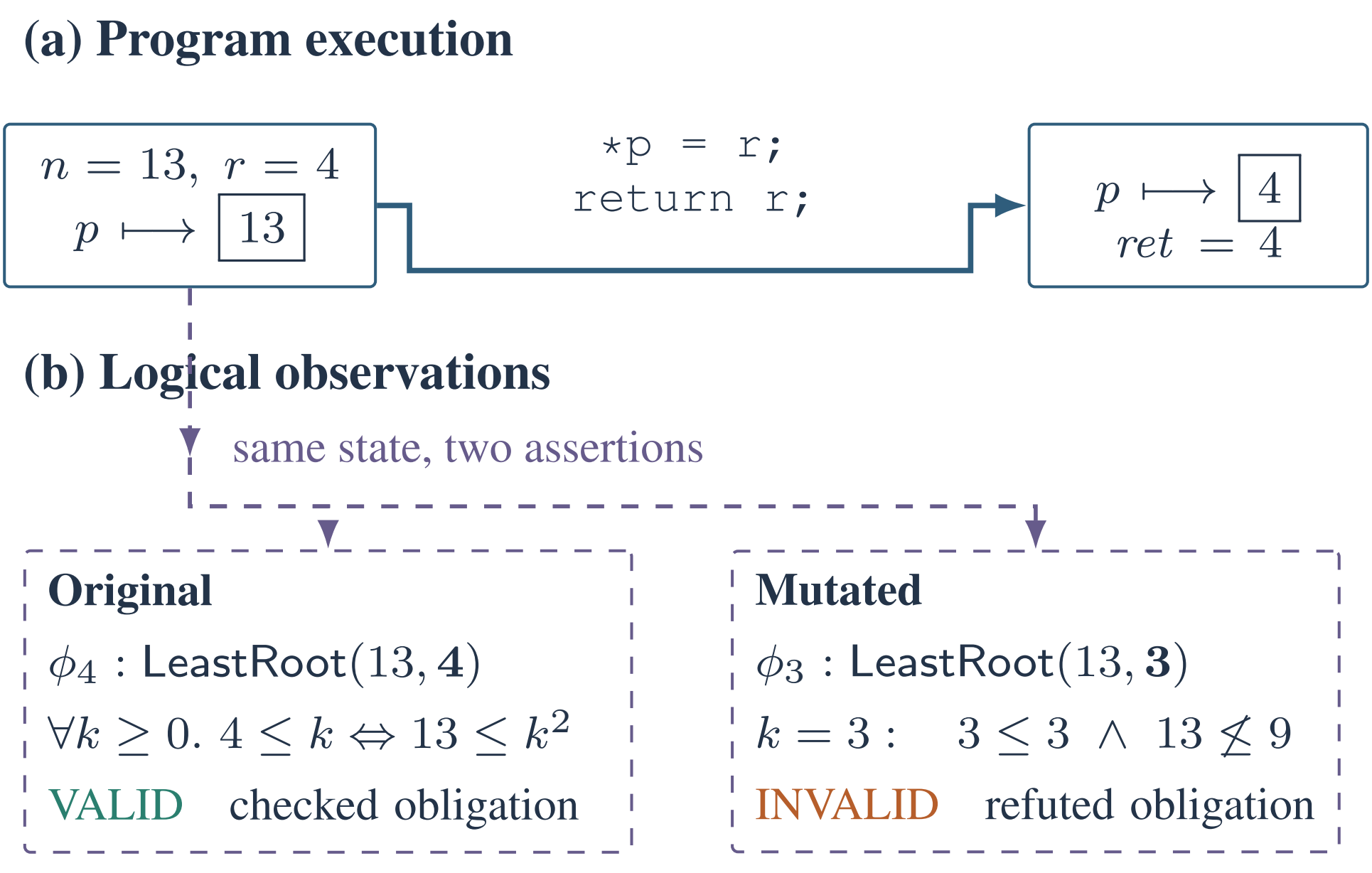

(a) Program execution
$n = 13,\ r = 4$
$p \longmapsto \boxed{13}$
*p = r;
return r;
$p \longmapsto \boxed{4}$
$ret = 4$
(b) Logical observations
same state, two assertions
Original
ϕ_4 : LeastRoot$(13, \mathbf{4})$
$\forall k \geq 0.\ 4 \leq k \Leftrightarrow 13 \leq k^2$
VALID checked obligation
Mutated
ϕ_3 : LeastRoot$(13, \mathbf{3})$
$k = 3 : \quad 3 \leq 3 \ \wedge \ 13 \not\leq 9$
INVALID refuted obligation